GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

Electronic Devices (EDC) — GATE Formula Reference

Semiconductor Physics, P-N Junctions, MOS Capacitor Regimes & MOSFET Equations

GATE Section: Section 3 • **Marks:** 9 - 11 Marks • **Core Focus:** Carrier Transport & Device Physics

ESSENTIAL SEMICONDUCTOR CONSTANTS & TRAPS

- **Thermal Voltage:** Vt = kT / q ~ 25.86 mV at 300 K (approx 26 mV in GATE).
- **Intrinsic Carrier Density:** ni for Silicon at 300 K ~ 1.5 x 10^10 cm^-3. Bandgap Eg(Si) = 1.12 eV.
- **Einstein Relation:** Dn / mu_n = Dp / mu_p = Vt = kT / q.

SECTION: CARRIER TRANSPORT & SEMICONDUCTOR PHYSICS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Mass Action Law	**n * p = ni^2**	ni^2 = Nc * Nv * exp(-Eg / kT)	Valid strictly in thermal equilibrium for non-degenerate semiconductors. Independent of doping!
Charge Neutrality	**n + Na^- = p + Nd^+** **n ~ Nd (n-type), p ~ Na (p-type)**	Na = acceptor doping, Nd = donor doping	Minority carriers: p_n = ni^2 / Nd (n-type), n_p = ni^2 / Na (p-type).
Total Current Density	**J = J_drift + J_diff** **J = q*(n*mu_n + p*mu_p)*E + q*Dn*(dn/dx) - q*Dp*(dp/dx)**	E = electric field Dn, Dp = diffusion coefficients	At thermal equilibrium with no external bias, J_total = 0 (drift and diffusion exactly cancel).
Einstein Relation	**Dn / mu_n = Dp / mu_p = Vt = kT / q**	Vt = 25.86 mV at 300 K	Connects diffusion coefficient directly to carrier drift mobility.
Hall Effect	**V_H = (I * B) / (q * n * w)** **R_H = 1 / (q * p) (p-type) or -1 / (q * n) (n-type)**	R_H = Hall coefficient w = sample thickness	Determines carrier type (sign of R_H), majority carrier concentration, and carrier mobility (mu = |R_H| * sigma).

SECTION: P-N JUNCTION & DIODES

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Built-in Potential	**V_bi = Vt * ln((Na * Nd) / ni^2)**	Vt = kT/q ~ 26 mV	Vbi increases logarithmically with doping. Decreases by ~2 mV/°C temperature increase.
Depletion Width	**W = sqrt([(2*epsilon/q) * (1/Na + 1/Nd) * (Vbi - V)])**	V = applied forward bias (positive) V = -VR for reverse bias	W is proportional to sqrt(Vbi + VR). Depletion layer penetrates deeper into the LIGHTLY doped side!
Maximum Electric Field	**E_max = 2 * (Vbi - V) / W** **E_max = (q * Nd * xn) / epsilon = (q * Na * xp) / epsilon**	Electric field is triangular, maximum at metallurgical junction	E_max is directly responsible for avalanche breakdown when it exceeds critical field E_crit.
Junction Capacitances	**C_dep = (epsilon * A) / W proportional to 1 / sqrt(Vbi + VR)** **C_diff = (tau_T * I_D) / (eta * Vt)**	C_dep dominant in reverse bias C_diff dominant in forward bias	Diffusion capacitance is directly proportional to forward diode current I_D!

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Shockley Diode Equation	**I = I0 * [exp(V / (eta * Vt)) - 1]**	I0 = reverse saturation current eta = ideality factor (1 for Si diffusion)	I0 doubles for every 10 °C rise in temperature. Diode forward voltage decreases by 2.5 mV/°C for constant current.

SECTION: MOSFET & MOS CAPACITOR PHYSICS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Flat-Band & Threshold Voltage	**V_FB = Phi_ms - (Q_ox / C_ox)** **V_T = V_FB + 2*phi_F + [sqrt(2*q*epsilon_s*Na*2*phi_F) / C_ox]**	phi_F = Vt * ln(Na / ni) = bulk Fermi potential C_ox = epsilon_ox / t_ox	Strong inversion begins when surface potential phi_s = 2 * phi_F.
Linear (Triode) Region	**I_D = mu_n * C_ox * (W/L) * [(V_GS - V_T)*V_DS - (V_DS^2 / 2)]**	Condition: V_GS >= V_T and V_DS < (V_GS - V_T)	For very small V_DS: MOSFET acts as voltage-controlled resistor with R_on = 1 / [mu_n*Cox*(W/L)*(Vgs - Vt)].
Saturation Region	**I_D = 0.5 * mu_n * C_ox * (W/L) * (V_GS - V_T)^2 * (1 + lambda*V_DS)**	Condition: V_GS >= V_T and V_DS >= (V_GS - V_T) lambda = channel length modulation parameter	Pinch-off occurs at drain end when V_DS = V_GS - V_T.
Transconductance	**g_m = dI_D / dV_GS | V_DS** **g_m = mu_n*C_ox*(W/L)*(V_GS - V_T) = sqrt(2*mu_n*C_ox*(W/L)*I_D) = 2*I_D / (V_GS - V_T)**	MOSFET small-signal transconductance	Output resistance: r_o = 1 / (lambda * I_D) = V_A / I_D.
Body Effect	**Delta V_T = gamma * [sqrt(2*phi_F + V_SB) - sqrt(2*phi_F)]**	gamma = body effect parameter = sqrt(2*q*epsilon_s*Na) / C_ox	Reverse source-to-body bias (V_SB > 0) INCREASES the threshold voltage V_T.